

B.Tech. Degree VII Semester Regular/Supplementary Examination November 2024

ME 19-205-0701 REFRIGERATION AND AIR CONDITIONING (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Understand the basics of different refrigeration cycles.
 CO2: Study the vapour compression and vapour absorption refrigeration cycles- both theoretical and practical cycles, properties of refrigerants and selection criteria for refrigerants.
 CO3: Understand the different components of refrigeration systems.
 CO4: Get knowledge of psychrometry, air-conditioning systems and will be able to do the cooling load calculation.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate,
 L6 – Create
 PO – Programme Outcome

PART A (Answer *ALL* questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a) Define the following:	3		L1	1	1
(i) Actual C.O.P.					
(ii) Theoretical C.O.P.					
(iii) Relative C.O.P.					
(b) What are the advantages and limitations of a vortex tube refrigeration system?	3		L2	1	1,2
(c) State merits and demerits of air refrigeration system.	3		L2	2	1,2
(d) What is the difference between wet compression and dry compression?	3		L1	2	1,2
(e) Name the different refrigerants generally used.	3		L1	3	1,2
(f) Explain the working of evaporative condensers.	3		L2	3	1,2
(g) State Dalton's law of partial pressure.	3		L2	4	1,2
(h) Define room sensible heat factor.	3		L1	4	1,2

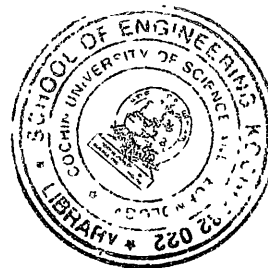
PART B

(4 × 12 = 48)

- II. (a) A Carnot refrigeration cycle absorbs heat at 270 K and rejects it at 300 K. 8 L3 1 1,2
- (i) Calculate the coefficient of performance of this refrigeration cycle.
- (ii) If the cycle is absorbing 1130 kJ/min at 270 K, how many kJ of work is required per second?
- (iii) If the Carnot heat pump operates between the same temperatures as the above refrigeration cycle, what is the coefficient of performance?
- (iv) How many kJ/min will the heat pump deliver at 300 K if it absorbs 1130 kJ/min at 270 K?
- (b) Differentiate clearly between open and closed air-refrigeration systems. 4 L2 1 1

OR

(P.T.O.)



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		Marks	BL	CO	PO																	
III.	(a) The temperature limits of an ammonia refrigerating system are 25°C and -10°C. If the gas is dry at the end of compression, calculate the coefficient of performance of the cycle assuming no sub cooling of the liquid ammonia.	8	L3	1	1,2																	
	(b) Explain the working of a bootstrap air cycle cooling system used for aircrafts.	4	L1	1	1																	
IV.	(a) 28 tonnes of ice from and at 0°C is produced per day in an ammonia refrigerator. The temperature range in the compressor is between 25°C and -15°C. The vapour is dry and saturated at the end of compression and an expansion valve is used. Assuming a co-efficient of performance of 62% of the theoretical, calculate the power required to drive the compressor. The following properties are given:	7	L3	2	1,2																	
<table><tr><th rowspan="2">Temp (°C)</th><th colspan="2">Enthalpy(kJ/kg)</th><th rowspan="2">Entropy of liquid (kJ/kg K)</th><th rowspan="2">Entropy of vapour (kJ/kg K)</th></tr><tr><th>Liquid</th><th>Vapour</th></tr><tr><td>25</td><td>100.04</td><td>1319.22</td><td>0.3473</td><td>4.4852</td></tr><tr><td>-15</td><td>-54.56</td><td>1304.99</td><td>-2.1338</td><td>5.0585</td></tr></table>						Temp (°C)	Enthalpy(kJ/kg)		Entropy of liquid (kJ/kg K)	Entropy of vapour (kJ/kg K)	Liquid	Vapour	25	100.04	1319.22	0.3473	4.4852	-15	-54.56	1304.99	-2.1338	5.0585
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	Liquid	Vapour																				
25	100.04	1319.22	0.3473	4.4852																		
-15	-54.56	1304.99	-2.1338	5.0585																		
	Take latent heat of ice = 335 kJ/kg.																					
	(b) Explain the effect of subcooling of condensate with the help of T-s and p-h diagrams in vapour compression systems.	5	L2	2	1,2																	
OR																						
V.	(a) Describe with a neat sketch the Li-Br and water vapour absorption refrigeration system. What are the limitations?	8	L2	2	1,2																	
	(b) What are secondary refrigerants? Why are these used in refrigeration and air conditioning industry? Name a few substances which are commonly used as secondary refrigerants.	4	L1	2	1,2																	
VI.	(a) What are the different types of rotary compressors? Explain the working principle of each one of them.	7	L2	3	1,2																	
	(b) Explain briefly the following air-cooled condensers:	5	L2	3	1,2																	
	(i) Natural convection condenser.																					
	(ii) Forced-air circulation condenser.																					
OR																						
VII.	(a) Explain with neat sketches the following evaporators:	8	L2	3	1,2																	
	(i) Flooded type evaporator.																					
	(ii) Dry expansion evaporator.																					
	(b) Discuss the operation of a water cooled condenser in a refrigeration system.	4	L2	3	1,2																	

(Continued)

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		Marks	BL	CO	PO
VIII.	(a) With the help of psychrometric chart, explain the following processes:	5	L1	4	1,2
	(i) Sensible heating and sensible cooling process.				
	(ii) Heating and dehumidification process.				
	(b) The sling psychrometer in a laboratory test recorded the following readings:	7	L3	4	1,2
	Dry bulb temperature = 35°C.				
	Wet bulb temperature = 25°C.				
	Calculate the following:				
	(i) Specific humidity.				
	(ii) Relative humidity.				
	(iii) Vapour density in air.				
	(iv) Dew point temperature.				
	(v) Enthalpy of air.				
	Take atmospheric pressure = 1.0132 bar.				
	OR				
IX.	(a) With the help of a neat sketch explain the main components of central air-conditioning system.	8	L2	4	1,2
	(b) 250 m ³ /min of air at atmospheric conditions 12°C DBT and 50% RH is supplied to an air-conditioned hall. The required conditions are 18°C DBT and 60% RH. Determine:	4	L3	4	1,2
	(i) Sensible heat and latent heat removed from the air per minute.				
	(ii) Sensible heat factor for the system.				

Bloom's Taxonomy Level's

L1 – 52.77%, L2 – 45.98%, L3 – 1.25%.

